Exp. no: 12a
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IMPLEMENT ECHO CLIENT SERVER USING TCP/UDP SOCKETS.

AIM:

To Implement echo client server using TCP/UDP sockets & demonstrate communication between a client & server.

PROCEDURE:

1) Open VS Code.
2) Create a folder
3) Create two Python files:
   server.py
   client.py
4) In server.py, create a socket using
   IPV4 addressing (AF_INET)
   TCP (SOCK_STREAM)
5) Bind the server socket to an IP address & port number.
6) Make the server listen for an incoming client connection
7) Accept the client connection.
8) Receive data from the client.
9) Display the received message
10) Send the same message back to the client.
11) In client.py, create a TCP socket.
12) Enter the server IP address.

Explorer

12a

client.py

server.py

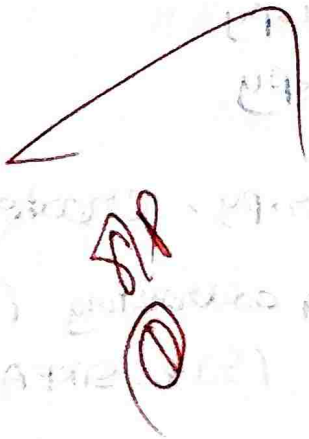
Welcome | server.py | client.py

server.py > ...

```
1   import socket
2   server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
3   server.bind(("0.0.0.0", 5000))
4   server.listen(1)
5   print("Server is waiting for connection...")
6   conn, addr = server.accept()
7   print("Connected to:", addr)
8   while True:
9       data = conn.recv(1024).decode()
10      if not data:
11          break
12      print("Client:", data)
13      conn.send(data.encode())
14  conn.close()
15  server.close()
```

Problems Output Debug Console Terminal Ports

```
PS D:\12a> python server.py
Server is waiting for connection...
```



13) Connect the client to the server
14) Enter a message from the client
15) Send the message to the server.
16) Receive the echoed response from the server.
17) Display the response
18) Repeat the communication for additional message.
19) Use exit to terminate the connection.

RESULT:

Thus, an echo client-server application was successfully implemented using socket programming.